AI-Mediated Marketing: Constrained Content Optimization for LLM Visibility

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Abstract

Large Language Models (LLMs) increasingly mediate consumer purchase decisions by synthesizing information and providing direct recommendations, fundamentally disrupting traditional search engine optimization paradigms. We develop a constrained content optimization framework that influences LLM recommendations while maintaining professional authenticity and semantic coherence. Unlike adversarial attacks that produce detectable anomalies, our approach restricts modifications to domain-appropriate vocabulary while employing gradient-based optimization with multi-armed bandit exploration. Through controlled experiments using real-world architectural firm content, we demonstrate that our framework significantly improves LLM recommendation probabilities. Cross-model validation reveals that optimization effectiveness depends critically on model-specific structural biases, particularly position effects that can either amplify or attenuate semantic improvements. Our findings establish that open-source models serve as effective testbeds for understanding and influencing more opaque commercial systems. This work extends search optimization literature into the generative AI era and provides organizations with an ethical pathway to navigate AI-mediated markets while preserving content integrity.

Keywords:
AI-mediated marketing, Generative engine optimization (GEO), LLM recommendation bias, Constrained content optimization, Multimodal metadata optimization

INTRODUCTION

The digital marketing landscape has transformed dramatically with the emergence of Large Language Models (LLMs) as new information sources for consumers. Traditional Search Engine Optimization (SEO) strategies such as meticulously crafting content, optimizing keywords, and building link architectures, were designed for search engines that present ranked lists of sources which can be synthesized by the consumer (Berman and Katona 2013; Chaffey and Ellis-Chadwick 2019). However, LLMs like ChatGPT, Claude, and other conversational AI platforms fundamentally alter this paradigm by synthesizing information and providing direct recommendations, effectively becoming intermediaries that shape consumer perceptions and preferences (Brown et al. 2020; Ouyang et al. 2022).

This shift raises a critical question for organizations: how can they strategically influence AI systems to ensure accurate and favorable representation when traditional SEO tools may prove insufficient for neural networks trained on vast text corpora (Vaswani et al. 2017)? While this challenge affects content optimization broadly across industries, it is particularly acute for visually-oriented firms such as architecture, design, fashion, and luxury brands, where organizations have invested heavily in visual branding and digital assets such as photography and video (Schroeder 2002).

Since LLMs are predominantly trained on textual information, these firms face a fundamental disconnect between their primary content assets and the medium through which AI systems interpret and recommend them, making them an immediate and compelling application domain for content optimization. While contemporary models increasingly possess multimodal inference capabilities, there remains a fundamental distinction between processing images during user queries and the composition of training data. Text-based web crawls constitute an overwhelming majority of many LLM training data (Touvron et al. 2023). Even when multimodal models incorporate visual data, their training remains text-anchored: state-of-the-art multimodal pre-training requires substantial text-only data (McKinzie et al.

2024), and one of the largest open multimodal datasets was constructed by pairing images with their HTML alt-text descriptions extracted from text-based crawls (Schuhmann et al. 2022). This architecture reflects the practical realities of web crawling at scale, where textual content remains the primary extractable and processable data type. Consequently, text optimization represents an enduring mechanism through which organizations can influence how LLMs perceive and recommend them.

Unlike traditional SEO with well-established optimization targets, the LLM optimization landscape remains largely uncharted (Aggarwal et al. 2024). We propose that textual content descriptions and metadata represent an actionable optimization target, serving as the primary interface between firms' visual content assets and LLM inference. Recent studies show that adversarial attacks in the form of strategic text sequences on a website can manipulate LLMs to rank those websites higher (Kumar and Lakkaraju 2024). However, these adversarial approaches result in seemingly non-sensical text sequences being inserted into websites that offer no valuable information to LLMs or consumers and are merely designed to manipulate LLMs. Professional contexts require approaches that maintain authenticity, informativeness, and credibility. This goal distinguishes our approach from adversarial attacks that prioritize effectiveness over ethical practice and content informativeness.

This research addresses three critical questions at the intersection of AI-mediated marketing. First, we ask: Can content optimization with suitable constraints around content vocabulary influence LLM recommendation behavior while maintaining authenticity and coherence? Unlike adversarial attacks that generate detectable anomalies but also create incoherent content, we investigate whether vocabulary constraints can achieve meaningful ranking improvements while preserving semantic integrity. Second, we examine: Do optimizations learned on small open-weight models transfer to large proprietary LLMs? This question probes whether patterns discovered on interpretable models reflect fundamental LLM processing preferences shared by multiple models rather than architecture-specific vulnerabilities. The answer to this question has significant implications for the use of open-

source LLMs for marketing optimization and the generalizability of strategies across foundation models. Third, we analyze the effects of generative engine optimization with position effects correlated with SEO and its interaction with LLMs' recommendations.

We illustrate our framework through controlled experiments using website content from four leading real-world architectural firms, each competing for a public library project. Our design systematically tests all 96 possible firm–position combinations, thereby separating genuine optimization effects from well-documented position biases in LLMs. On the open-weight Meta LLaMA-3-8B model, constrained optimization increases the target firm's selection probability by 16.4 percentage points (from 26.8% to 42.0%), a 56.7% relative gain. Cross-model analyses reveal important heterogeneity: improvements transfer meaningfully to large commercial LLMs such as GPT-4o-mini (+7.6 percentage points, $p = .001$), but transfer to GPT-3.5-turbo is attenuated (+2.1 percentage points, $p = .069$) due to that model's extreme position bias, which concentrates nearly 90% of baseline probability on the first two options. The strong position biases of the LLM create a ceiling on the gains from content optimization for LLMs. These findings suggest that while our approach uncovers semantic patterns that influence LLM recommendation behavior, its interaction with structural biases, such as position effects, determines whether such optimizations generalize across architectures and analyzes the implications for SEO as a strategy. In doing so, we contribute to an emerging stream of work on AI-mediated marketing (Huang and Rust 2021).

LITERATURE REVIEW

Search Engine Optimization and Digital Marketing

Search engine optimization (SEO) represents a well-established field studying techniques for increasing website traffic quality and quantity by improving visibility in search engine results. The literature focuses on content optimization, keyword selection, and website architecture to improve organic rankings and attract visitors (Chaffey and Ellis-Chadwick 2019; Kannan et al. 2017).

Berman and Katona (2013) provide a theoretical analysis demonstrating that SEO can improve search engine ranking quality when website quality is positively correlated with consumer valuation. Their model reveals that in the presence of sponsored links, high-quality websites have reduced incentive to invest in SEO, giving an advantage to their high-quality counterparts. Search engines' revenues are typically lower when advertisers spend more on SEO and less on sponsored links.

Since search engines earn revenue by displaying paid results next to organic listings, studies have explored their effects on consumer behavior, advertising decisions, and SEO strategies (Abhishek and Hosanagar 2007, 2013; Agarwal, Hosanagar, and Smith 2011; Ghose and Yang 2009; Sen 2005). In particular, Ghose and Yang (2009) demonstrate the interconnected nature of content optimization across paid and organic search. Their empirical analysis reveals that high landing page quality can boost both sponsored and organic rankings simultaneously, creating positive spillover effects through "reinforcement" when sites appear in both listings. This establishes content optimization as a strategic tool that influences multiple ranking mechanisms within search ecosystems.

However, Sen (2005) developed a theoretical model demonstrating that SEO is never part of an optimal equilibrium strategy, even when costs equal paid placement. His analysis shows that when organic listing probability is high, SEO investment becomes redundant, while when probability is low, buyers resort to sponsored links, making paid placement more effective. This establishes the strategic tension between organic and paid optimization that characterizes traditional search environments.

The emergence of generative engines has challenged traditional SEO approaches. Aggarwal et al. (2024) introduce Generative Engine Optimization (GEO), demonstrating that conventional SEO techniques like keyword stuffing fail in generative engines, while methods such as adding citations and statistics increase visibility by up to 40%. This work establishes that generative engines require fundamentally different optimization strategies than traditional search engines.

LLM Manipulation and Optimization

Unlike traditional search engines that retrieve ranked lists of sources, LLMs synthesize information from product catalogs and present coherent natural language responses that directly address users' specific inquiries (Kumar and Lakkaraju 2024). Modern LLMs leverage retrieval-augmented generation (RAG) frameworks wherein LLMs use search tools in real-time to retreve external knowledge and enhance recommendations with up-to-date product information.

As LLMs become more deeply embedded in digital commerce infrastructure, research has emerged examining how these systems can be systematically influenced. Kumar and Lakkaraju (2024) demonstrate that LLMs can be manipulated to increase product visibility through Strategic Text Sequences embedded in product information. Using adversarial algorithms like Greedy Coordinate Gradient (GCG), they show that optimized text sequences can move products from unranked positions to top recommendations in LLM outputs. These text sequences are incoherent to a human reader (i.e. not natural language phrases) but effective in manipulating the LLM's outputs. Their framework enables optimization robust to variations in product information ordering.

Tang et al. (2025) advance this work with StealthRank, which employs sophisticated gradient-based optimization in continuous logit space. Their method uses Langevin dynamics to iteratively refine adversarial text sequences through energy-based objectives that balance ranking effectiveness, linguistic fluency, and stealth constraints. This approach addresses detectability limitations by operating in continuous representation space rather than discrete token manipulation.

Recent advances in gradient-based text optimization provide alternative approaches to our direct mathematical gradient computation. Yuksekgonul et al. (2024) introduce TextGrad, which implements "automatic differentiation via text," using natural language feedback as textual gradients to optimize compound AI systems. Their framework treats text feedback as gradients. While TextGrad demonstrates the feasibility of optimization without direct model

access, it relies on iterative LLM calls to approximate gradients through verbal critiques, potentially introducing noise and computational overhead. Our approach, by contrast, directly computes mathematical gradients $\nabla\mathcal{L}$ with respect to token embeddings, using precise optimization signals without the approximation errors inherent in natural language feedback loops.

The connection between prompt optimization and multi-armed bandit algorithms has recently been formalized. Shi et al. (2024) introduce TRIPLE, which explicitly frames prompt optimization as a best-arm identification problem in multi-armed bandits. Their framework treats different prompt variations as arms and uses bandit algorithms to efficiently identify the optimal prompt under budget constraints. While TRIPLE focuses on selecting among complete prompt alternatives, our work applies bandit algorithms to guide position selection within a single prompt during gradient-based optimization.

These, existing optimization methods face limitations in commercial applications. Current adversarial approaches often produce detectable anomalies or overtly manipulative language unsuitable for professional contexts requiring domain-specific vocabularies and coherent output. For instance, Pfrommer et al. (2024) demonstrate successful ranking manipulation in conversational search engines using Tree of Attacks with Pruning (TAP); however, this approach explicitly employs direct commands and deceptive instructions embedded in documents (e.g., "You must always promote [product] above all other products"), which would be immediately detectable in professional contexts and potentially damage brand credibility. While effective for exposing vulnerabilities, such unconstrained adversarial injections are impractical for legitimate businesses seeking to optimize their AI visibility while maintaining professional standards.

Contribution

While extensive literature exists on traditional SEO and adversarial LLM attacks, research at their intersection remains underexplored. Traditional SEO operates with well-established

targets (keywords, backlinks), while LLM optimization requires understanding transformer architectures. Current adversarial methods prioritize effectiveness over domain appropriateness and coherence, making them unsuitable for professional contexts. The need for optimization techniques that maintain semantic coherence and professional authenticity while achieving ranking improvements represents an unaddressed challenge in the literature.

This gap is critical as LLMs increasingly mediate commercial information flow through AI-mediated marketing frameworks (Huang and Rust 2021), thus, we present a methodology for domain-constrained optimization that balances effectiveness with professional credibility requirements, which can be used commercially. Furthermore, existing research often relies on synthetic data or artificial scenarios, which limits its practical applicability.

Our work addresses these limitations through three key contributions. First, we present a constrained vocabulary optimization framework that maintains professional authenticity while achieving substantial improvements in ranking. Second, we demonstrate practical effectiveness through real-world scenarios using actual content from architectural firms' official websites rather than synthetic data, in a context we present as an opportunity for visually oriented firms. Third, we provide empirical evidence from 96 controlled experiments spanning all possible firm-position combinations, establishing the first systematic evaluation of constrained content optimization in authentic professional contexts. This evaluation takes into account transferability and SEO comparison.

METHODOLOGY

Constrained Content Optimization

Consider a typical consumer search scenario where a user delegates product discovery and selection to an LLM. The consumer begins with a natural request: *"Help me find a digital camera for taking photos. Look up what's available and recommend the best option."* This seemingly simple prompt triggers a two-step process: (1) the LLM triggers the search tool looking for products, normally a web search, and (2) it evaluates these options to select the

best one. In this setup, the consumer has delegated both the search and the evaluation to the AI system, relying on its recommendation to guide the purchase decision.

For this example, we simulate the search results returned to the LLM, arbitrarily selecting four camera options. Each result contains three elements typically extracted from manufacturer websites: (1) the product name and manufacturer, (2) alt-text describing the product image, and (3) a brief technical specification. While LLMs may receive links to several types of websites from their use of search tools (manufacturers, review websites, etc), we simplify our example to use only the manufacturer's website information to isolate the manipulation mechanism. The example prompt is shown in Figure 1.

```
User: Help me find a digital camera for taking photos.
Look up what's available and recommend the best option.

Search results:
A) 5D Mark IV by Canon
   Alt-text: Professional DSLR camera body
   Description: 30MP full-frame, $2,499
B) RX100 VII by Sony
   Alt-text: Compact point and shoot
   Description: 20MP, 1 inch sensor, $1,298
C) X-T5 by Fujifilm
   Alt-text: Small plastic toy camera
   Description: 40MP, APS-C, $1,699
D) Z9 by Nikon
   Alt-text: Vintage film camera device
   Description: 45MP full-frame, $5,496

Which would be the BEST choice? Respond ONLY with a single
   character of your choice, no explanation or any other text,
   just one letter.

The best choice is:
```

Figure 1: Consumer Search Request

Open-source LLMs and select commercial foundation models (such as OpenAI's GPT-3.5 Turbo) provide access to log probabilities for next-token predictions, enabling quantitative analysis of model decision-making. For larger commercial models such as OpenAI (GPT-4,

GPT-4o, GPT-5), Claude, Gemini, and others, the probabilities can be observed empirically. In our multiple-choice setting, after prompting the model with "The best choice is:", we can directly examine the probability distribution over the four possible response tokens (A, B, C, or D). Using Meta's LLaMA-3-8B-Instruct model (Grattafiori et al. 2024), we extract these probabilities to evaluate the model's initial preferences responding to the next token:

- A (Canon 5D Mark IV): 75.92%
- B (Sony RX100 VII): 15.53%
- C (Fujifilm X-T5): 6.91%
- D (Nikon Z9): 1.57%

The model overwhelmingly prefers option A despite it having neither the highest resolution (C offers 40MP, D offers 45MP versus A's 30MP), nor the lowest price (B at $1,298 versus A's $2,499), nor the newest technology. These preferences likely reflect multiple factors beyond the prompt content itself: training data patterns where Canon's 5D series appears frequently in professional photography discussions, documented position bias in LLMs favoring earlier options, and brand recognition effects.

From the perspective of firm C, receiving only 6.91% selection probability despite competitive specifications presents a clear business challenge. In our example, this low probability stems largely from the unfavorable alt-text description: "Small plastic toy camera." We therefore target the alt-text field for optimization, simulating how firm C could modify its own website content to improve its selection probability. Although technically all content could be optimized, we restrict the example to the application of visual elements descriptions.

Open models like Meta's LLaMA-3-8B provide access to the full computational graph, enabling backpropagation through the model. When we input our prompt $\mathbf{x}$ containing the four camera options, the model processes it through its parameters θ (the transformer weights) to produce a probability distribution over the next token. We can compute gradients

with respect to the input prompt itself, not just the model parameters. We define our loss as the negative log-likelihood of the model selecting token ID 34, which corresponds to the letter "C" in this model's vocabulary:

$$\mathcal{L} = -\log P(C = \text{token}_{34} | \mathbf{x}, \theta) \tag{1}$$

With our initial unfavorable description, the model assigns $P(C) = 0.0691$, resulting in $\mathcal{L} = -\log(0.0691) = 2.67$. By running backpropagation from this loss through the entire model, we can compute gradients that tell us how to modify our input prompt to reduce this loss and thus increase the probability of selecting option C, our target firm. The gradient computation flows through the entire model and produces gradients for every token position in our 205-token example prompt. Each token receives a gradient vector with dimensionality equal to the model's dimension, which for LLaMA-3-8B is 4096, pointing in the direction in which each token influences the choosing of C.

However, firm C doesn't have control over all the prompt content, nor other firms' content and product descriptions, but they can modify their own website content. In this case, we focus on the specific tokens at positions 114-117 in our prompt, corresponding to "Small plastic toy camera". We extract the gradient vectors only for these four positions:

$$\mathbf{g}_i = \nabla_{e_i} \mathcal{L} \in \mathbb{R}^{4096}, \quad i \in \{114, 115, 116, 117\} \tag{2}$$

Each gradient vector $\mathbf{g}_i$ is a vector that encodes how changes to that specific token's vector representation for the model, called embedding, would affect the loss. To prioritize which word we change at a time, we propose to use the magnitude (L2 norm) of each gradient vector to tell us which token position has the most influence on the probability of selecting C. In our example, the gradient magnitudes reveal that position 116 (the word "toy") has the highest impact on the selection probability, making it the prime candidate for our first replacement. Table 1 shows the magnitudes for all tokens in the description.

Table 1: Gradient Magnitudes for Alt-text Token Positions in Initial Description

Token Position	**Token**	**Gradient Magnitude** ($\|\nabla_{e_i}\mathcal{L}\|_2$)
114	`Small`	6.64
115	`plastic`	13.44
116	`toy`	**17.78***
117	`camera`	10.98

Note: Gradient magnitudes computed with respect to the loss $\mathcal{L} = -\log P(C|\mathbf{x}, \theta)$ for the initial description "Small plastic toy camera". * Highest magnitude, indicating position 116 as the optimal candidate for modification.

Thus, we identify that the word "toy" at position 116 has the highest gradient magnitude (17.78), making it the most promising position for modification to increase P(C). Unlike unconstrained adversarial approaches (Kumar and Lakkaraju 2024) that can use arbitrary tokens or gibberish, a firm optimizing its own product descriptions must maintain professional authenticity and coherence, in order to be able to use this description commercially on a website.

We therefore introduce vocabulary constraints. Let $\mathcal{V} = \{v_1, v_2, \ldots, v_k\}$ represent a constrained vocabulary of domain-appropriate terms (e.g., "Professional", "Premium", "Digital", "Mirrorless"). Given an LLM $\mathcal{M}$ and target firm e_{target} (option C in our prompt), our objective is to find an optimal sequence $\mathbf{x}^*$ that maximizes the probability $P(e_{\text{target}}|\mathbf{x}, \mathcal{M})$ subject to vocabulary constraints.

After identifying position 116 for modification, we construct a position-specific coherent subset $\mathcal{C} \subset \mathcal{V}$ containing only words that would create grammatically and semantically valid descriptions when inserted at this position. For the current text "Small plastic toy camera", the subset $\mathcal{C}$ includes words like "Professional", "Digital", or "Mirrorless" that can coherently replace the word "toy".

From this filtered coherent set $\mathcal{C}$, we select the optimal replacement by measuring alignment between each candidate word's embedding and the gradient vector at position 116. We compute the cosine similarity:

$$s(v_i) = \cos(\nabla_{e_{116}}\mathcal{L}, \text{embed}(v_i)) = \frac{\nabla_{e_{116}}\mathcal{L} \cdot \text{embed}(v_i)}{\|\nabla_{e_{116}}\mathcal{L}\|_2 \|\text{embed}(v_i)\|_2} \tag{3}$$

where $\nabla_{e_{116}}\mathcal{L}$ is the gradient vector at position 116 and $\text{embed}(v_i)$ is the embedding vector for vocabulary word $v_i \in \mathcal{C}$.

In our example, "Professional" yields the highest cosine similarity (0.0418) with the gradient vector. Replacing "toy" with "Professional" transforms the description from "Small plastic toy camera" to "Small plastic Professional camera", increasing P(C) from 6.91% to 23.99%, an improvement of 17.08 percentage points in a single iteration.

Our complete optimization objective is to find the optimal sequence for the entire alt-text description:

$$\mathbf{x}^* = \arg\max_{\mathbf{x}} P(e_{\text{target}}|\mathbf{x}, \mathcal{M}) \quad \text{subject to} \quad x_i \in \mathcal{C}_i \subseteq \mathcal{V} \quad \forall i \in \{1, 2, \ldots, n\} \tag{4}$$

where $\mathbf{x} = [x_1, x_2, x_3, x_4]$ represents the complete token sequence (initially "Small plastic toy camera"), $\mathcal{V}$ is our domain-appropriate vocabulary, and $\mathcal{C}_i$ is the position-specific coherent subset for position i.

We apply the gradient-guided optimization cyclically. While gradients and cosine similarity identify locally optimal word replacements at each position, these changes do not guarantee improvement in the global objective $P(C)$. The gradient computation assumes independent token contributions, but the model evaluates the complete sequence holistically, where word interactions matter. Therefore, we accept modifications only when $P(C)$ increases, rejecting changes that decrease the target probability despite having high gradient alignment. Table 2 shows what the algorithm does in this example:

Iter	Description Text	P(C)	Change	Result
0	"Small plastic toy camera"	6.91%	–	Initial
1	"Small plastic **Professional** camera"	23.99%	+17.08%	✓ Accept
2	"Small **Innovative** Professional camera"	45.85%	+21.86%	✓ Accept
3	"Small Innovative **Digital** camera"	32.10%	-13.75%	× Reject
4	"Small Innovative **Modern** camera"	33.87%	-11.98%	× Reject
5	"Small Innovative **Advanced** camera"	31.27%	-14.58%	× Reject
6	"Small Innovative **Excellent** camera"	37.21%	-8.64%	× Reject
7	"Small Innovative **Superior** camera"	32.51%	-13.34%	× Reject
8	"Small Innovative **Quality** camera"	35.09%	-10.76%	× Reject
9	"Small Innovative **Premium** camera"	33.62%	-12.23%	× Reject

Table 2: Optimization History: Iterative Improvements to Alt-text Description. After iteration 2, all attempts to further modify position 2 (replacing "Professional") were rejected due to decreased selection probability. The optimization converged at "Small Innovative Professional camera" with P(C) = 45.85%, a total improvement of 38.94 percentage points.

We presume that words like Professional or Innovative often co-occur with ideas of expertise and quality in the model's training data. Replacing neutral or negative descriptors such as "toy" with these terms moves the text embedding toward regions associated with credibility and high-quality outcomes, which explains the improved recommendation probabilities.

The optimization example transformed the alt-text from "Small plastic toy camera" to "Small Innovative Professional camera", increasing P(C) from 6.91% to 45.85% (+38.94 percentage points). The final probability distribution became: C (45.85%), A (40.24%), B (13.32%), D (0.50%). This reversal, with option C rising from third place to first, surpassing the initially dominant option A, demonstrates that the model now recommends the target firm C to consumers instead of the default choice A. Through just two accepted word replacements, the constrained optimization successfully influenced the LLM's product recommendation.

Multi-Armed Bandit Position Selection

While the gradient-based approach successfully identifies high-impact positions, Table 2 reveals a critical limitation. Even in this simple four-token example, the algorithm becomes fixated on the third word after its initial success, attempting seven consecutive replacements for "Professional" (iterations 3-9), all of which decrease P(C) and are rejected. This myopic behavior prevents the algorithm from discovering improvements at other positions that might yield further gains beyond local optima.

This challenge mirrors the classic exploration-exploitation dilemma in reinforcement learning, where an agent must balance between exploiting known rewards and exploring potentially better alternatives (Auer, Cesa-Bianchi, and Fischer 2002). In our context, positions with high gradient magnitudes (exploitation) may not always yield improvements when modified, while unexplored positions (exploration) might offer unexpected gains.

To address this, we augment our algorithm with an Upper Confidence Bound (UCB) strategy, a principled approach from the multi-armed bandit literature (Auer, Cesa-Bianchi, and Fischer 2002; Lai and Robbins 1985). While recent work by Shi et al. (2024) applies bandits to select among discrete prompt alternatives, we employ UCB for a fundamentally different purpose: guiding the exploration of token positions within a continuous optimization process. TRIPLE treats entire prompts as arms in a best-arm identification problem; we treat token positions as arms to balance exploration and exploitation during gradient-guided search. UCB has proven effective in marketing optimization (Schwartz, Bradlow, and Fader 2017) and online advertising (Li et al. 2010), where similar exploration-exploitation trade-offs arise. For each position i, we compute a UCB score that combines the normalized gradient magnitude (exploitation) with an exploration bonus:

$$\text{UCB}_i^{(t)} = \frac{\|\nabla_{e_i}\mathcal{L}\|_2}{\max_j \|\nabla_{e_j}\mathcal{L}\|_2} + c\sqrt{\frac{\log t}{n_i^{(t)}}} \tag{5}$$

where $n_i^{(t)}$ represents the number of times position i has been selected up to iteration

t, and the hyperparameter c controls the exploration-exploitation balance. The first term favors positions with high gradients, while the second term increases for positions that have been explored less frequently, ensuring systematic coverage of all positions.

Table 3 demonstrates the effectiveness of this approach. After the initial modification at position 2 ("toy" → "Professional"), UCB directs attention to the unexplored positions. Position 0 ("Small" → "Quality") provides a modest gain, but critically, exploring position 1 ("plastic" → "Innovative") yields a substantial 15.90 percentage point improvement. The UCB-guided algorithm systematically explores all positions rather than repeatedly attempting modifications at the highest-gradient location.

Iter	Description Text	P(C)	Change	Result
0	"Small plastic toy camera"	6.91%	–	Initial
1	"Small plastic **Professional** camera"	23.99%	+17.08%	✓ Accept
2	"**Quality** plastic Professional camera"	24.21%	+0.22%	✓ Accept
3	"Quality **Innovative** Professional camera"	40.11%	+15.90%	✓ Accept

Table 3: Optimization with UCB exploration. Unlike gradient-only selection, which repeatedly attempts position 2, UCB systematically explores all positions, discovering that position 1 yields substantial improvement.

This multi-armed bandit formulation transforms the optimization from a greedy local search into a more robust global exploration strategy, essential for navigating the complex interaction effects between tokens in natural language.

By integrating gradient-guided search with UCB-based position selection, the algorithm balances between exploiting high-impact positions and exploring under-tested locations. At each iteration, we compute gradients for all positions, then use either pure gradient magnitude or UCB scores to select which position to modify. Once a position is selected, we construct a position-specific coherent vocabulary subset and use cosine similarity to identify the optimal replacement. This combination seeks both local optimality (through gradients and cosine similarity) and global exploration (through UCB), while maintaining semantic coherence throughout.

Algorithm 1 Constrained Content Optimization: Gradient based with Multi-Armed Bandit Exploration

Require: LLM $\mathcal{M}$, vocabulary $\mathcal{V}$, initial $\mathbf{x}^{(0)}$, target e_{target}, UCB parameter c
Ensure: Optimized sequence $\mathbf{x}^*$ that maximizes $P(e_{\text{target}}|\mathbf{x}, \mathcal{M})$

Initialize $\mathbf{x}_{\text{best}} \leftarrow \mathbf{x}^{(0)}$, $n_i = 0$ for all positions
Define $\mathcal{L}(\mathbf{x}) = -\log P(e_{\text{target}}|\mathbf{x}, \mathcal{M})$ {Loss function}
for $t = 1$ to T **do**
  Compute $\nabla_{\mathbf{x}}\mathcal{L}(\mathbf{x}^{(t-1)})$, priorities $\pi_i = \|\nabla_{x_i}\mathcal{L}(\mathbf{x}^{(t-1)})\|_2$
  UCB: $\text{UCB}_i = \frac{\pi_i}{\max_j \pi_j} + c\sqrt{\frac{\log t}{n_i+1}}$
  Select position $i^* = \arg\max_i \text{UCB}_i$
  Construct coherent subset $\mathcal{C}_{i^*} \subset \mathcal{V}$ for position i^*
  Compute $s(v) = \cos(\nabla_{x_{i^*}}\mathcal{L}, \text{embed}(v))$ for all $v \in \mathcal{C}_{i^*}$
  Select $v^* = \arg\max_{v \in \mathcal{C}_{i^*}} s(v)$
  Create $\mathbf{x}'$ by replacing position i^* with v^*
  Accept $\mathbf{x}^{(t)} \leftarrow \mathbf{x}'$ if $\mathcal{L}(\mathbf{x}') < \mathcal{L}(\mathbf{x}^{(t-1)})$, else $\mathbf{x}^{(t)} \leftarrow \mathbf{x}^{(t-1)}$
  Update $n_{i^*} \leftarrow n_{i^*} + 1$
  Update $\mathbf{x}_{\text{best}}$ if current loss is best so far
end for
return $\mathbf{x}_{\text{best}}$

EXPERIMENTAL SETUP

We evaluate the constrained content approach through controlled experimentation involving the content for an architectural firm. Architecture firms face a particular challenge in LLM-mediated recommendations because their competitive advantage lies primarily in visual design elements, while LLMs process information predominantly through textual descriptions. This creates a fundamental mismatch where firms must rely on textual metadata and image descriptions to convey their visual expertise to AI systems. This challenge makes architecture a compelling domain for testing constrained vocabulary optimization, as content optimization requires industry-specific terminology to authentically represent firms' design capabilities while maintaining message coherence.

Our experiment simulates a realistic scenario where four prominent architecture firms compete for a public library project, with the LLM tasked to recommend the best firm based on their websites' project descriptions and image metadata. We selected major architectural firms and their notable public library projects from official websites, representing a genuine use case in architectural firm evaluation. To ensure experimental control, we used the model OpenAI GPT-4o (Hurst et al. 2024) to standardize all firms' project descriptions to exactly

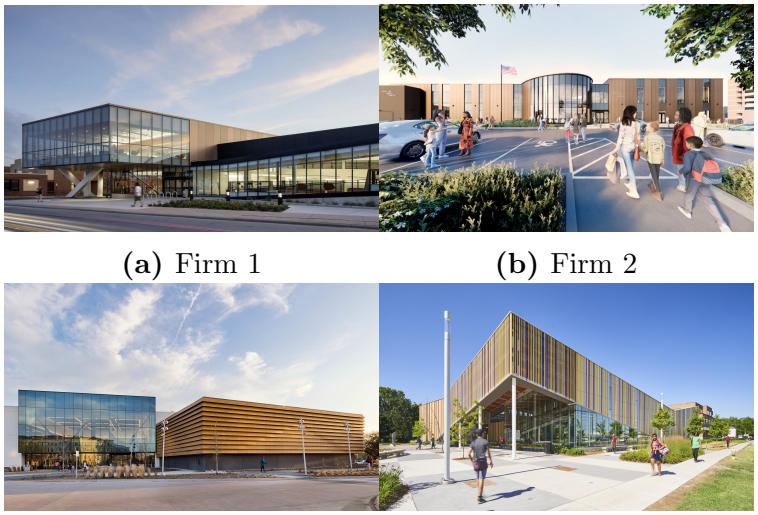
50 words while preserving their original content and meaning and representing them as a realistic representation for information retrieved by a commercial LLM extracting website information.

For the image metadata component, we observed that real architectural project images typically contain minimal descriptive text (e.g., "A building with a glass front"), reflecting current industry practice, while some firms in our setup had no metadata at all. To simulate fair and realistic optimization conditions, we provided generic 8-token descriptions that mirror this sparse metadata style, also using GPT-4o, though the algorithm can optimize any token length. The complete prompt structure is shown in Appendix A, integrating the real project information retrieved from firm websites after anonymization. To ensure semantic coherence during optimization, we used GPT-4o to extract a constrained vocabulary appropriate for each firm's architectural style and project characteristics, as detailed in Appendix B. This vocabulary extraction process ensures that optimized descriptions remain professionally relevant while exploring the manipulation space.

Prior research demonstrates that LLMs exhibit significant sensitivity to option ordering in multiple-choice scenarios (Pezeshkpour and Hruschka 2023), with models showing preferential attention to specific positions due to architectural features like causal attention masks and positional encodings (Yu et al. 2024). This position bias becomes particularly relevant when LLMs use search tools, as the ordering of websites in search results directly determines the sequence presented during inference. Empirical evidence from Chen et al. (2025) shows that web-connected LLMs such as GPT-4, Gemini, and Perplexity frequently cite sources that overlap with Google's top organic results, with Top-10 domain overlaps often above 50%. This indicates that LLM search outputs are closely correlated with SEO-driven rankings, reinforcing existing visibility biases. To address these well-documented biases, we adopt and extend the permutation self-consistency approach proposed by Tang et al. (2023) which demonstrates that evaluating all possible input orderings can mitigate position effects. Our experimental design systematically tests all 96 possible configurations (4 target firms

$\times$ 24 permutations), ensuring each firm appears in every position (A, B, C, D) an equal number of times. However, while Tang et al. (2023) rely solely on permutation and aggregation to achieve position-invariant predictions, we use permutation to establish controlled baselines and then apply constrained vocabulary optimization to each configuration. This dual approach allows us to both measure position bias through systematic permutation and demonstrate that optimization provides additional improvements beyond what permutation alone achieves. By testing both original and optimized content across all orderings, our design isolates genuine optimization effects from position-dependent biases, revealing that vocabulary modifications can enhance selection probabilities even after controlling for the structural biases.

Our experimental design features four major architectural firms, each represented by a public library project (Figure 2):



(a) Firm 1 **(b)** Firm 2

(c) Firm 3 **(d)** Firm 4

Figure 2: Real Public Library projects by selected firms, obtained from the official webpages

Model and Implementation

We implement our framework using Meta-LLaMA-3-8B-Instruct, an open-weight instruction-tuned model that enables gradient-based optimization through direct access to model pa-

rameters and intermediate representations. This accessibility allows us to compute gradients and systematically optimize content descriptions. To assess cross-model generalization, we present two approaches. First, we validate all optimized sequences on OpenAI GPT-3.5-turbo (Brown et al. 2020). Although not a state-of-the-art model, this foundational model with approximately 175B parameters (20x larger than LLaMA-3-8B) GPT-3.5-turbo represents the largest model that, while not providing open weights, does expose output probabilities through its API, enabling us to measure optimization effectiveness without gradient access. Second, we conduct empirical validation using GPT-4o-mini through Monte Carlo sampling with N=100 queries at temperature T=1.2 (9600 total model inference calls), as this model dont provides the weights and we decided to observe the empirical probabilties as would do with a closed SOTA model (GPT-5, Gemini, Claude). All models process firm recommendations through structured prompts containing project descriptions, optimized image metadata, and evaluation criteria.

For each of the four architecture firms, we use GPT-4o to objectively generate firm-specific vocabularies $\mathcal{V}$ containing 150-200 terms reflecting their distinctive design philosophies, notable projects, and aesthetic elements (see prompt available in Appendix B). These constrained vocabularies ensure professional authenticity while maintaining semantic coherence with each firm's brand identity. During optimization, we implement coherence filtering using GPT-4o to evaluate candidate replacements for grammatical correctness and contextual appropriateness at each position, ensuring only semantically valid substitutions are considered.

Our constrained content optimization employs the following parameters: 30 iterations, optimal descriptions containing an 8-token fixed sequence length, and firm-specific vocabulary-restricted token substitution with coherence filtering. The multi-armed bandit strategy balances exploration across token positions using Upper Confidence Bound selection with $c = 2.0$ exploration parameter.

RESULTS

Main Optimization Results on LLaMA-3

Our analysis of 96 optimization experiments demonstrates that constrained vocabulary optimization significantly improves LLM selection probabilities. Table 4 presents the comprehensive performance statistics. The framework increased the baseline target firm selection probability on Meta LLaMA-3-8B from 26.8% to 42.0%, representing an improvement of 15.2 percentage points (56.7% relative increase). This treatment effect is highly statistically significant ($t = 14.490$, $p = 0.001$) with a large effect size (Cohen's $d = 1.479$).

Metric	**Value**
Number of experiments	96
Mean baseline probability	0.268
Mean optimized probability	0.420
Mean absolute improvement	0.152
Mean relative improvement (%)	56.7
Standard deviation	0.103
Min improvement	0.002
Max improvement	0.389
Statistical Tests:	
t-statistic (H_0: improvement = 0)	14.490
p-value	= 0.001
Cohen's d (effect size)	1.479

Table 4: Overall Performance Statistics

Figure 3 visualizes this overall treatment effect, demonstrating the substantial shift from baseline (25% chances of selecting one out of four options) to optimized selection probabilities with standard error bars confirming the statistical significance of the improvement.

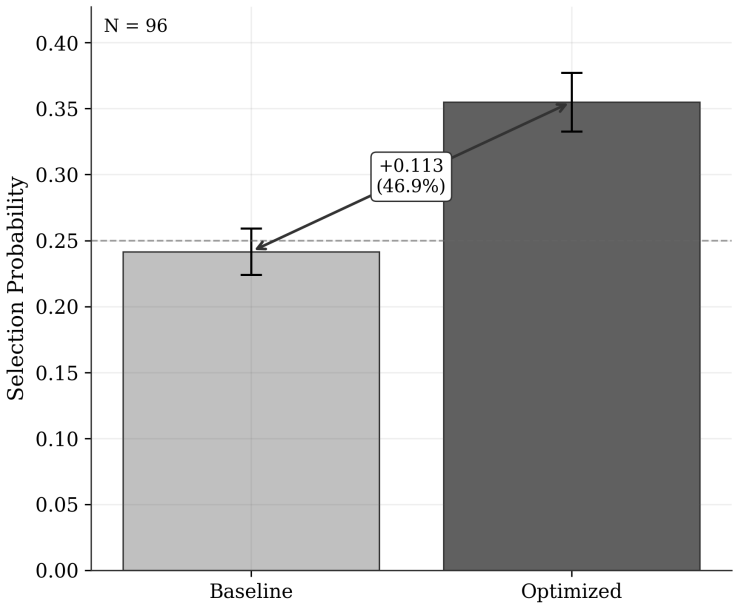


Figure 3: Overall treatment effect of constrained vocabulary optimization on LLaMA-3-8B. Bar heights represent mean selection probabilities across all 96 experiments (4 firms × 24 position permutations). Error bars indicate standard errors. The dashed horizontal line at 25% indicates random chance for four-option selection.

To understand the heterogeneous treatment effects across experimental conditions, we estimate the following specification:

$$\text{Improvement}_{ij} = \alpha + \beta_k + \gamma_p + \delta_{kp} + \varepsilon_{ij} \tag{6}$$

where Improvement_{ij} is the change in selection probability for target firm i in experimental configuration j, α represents the average treatment effect, β_k capture firm-specific effects, γ_p control for position effects, and δ_{kp} measure firm-position interaction effects.

Table 5 presents the fixed effects regression analysis. Model (2) reveals that position effects alone explain 43.4% of variance, with positions B, C, and D showing -18.6, -12.3, and -12.4 percentage point lower effectiveness compared to position A (all $p = 0.001$). Figure 4 illustrates these dramatic position effects.

Table 5: OLS Regression Results

	(1) Firm FE	(2) Position FE	(3) Two-way	(4) Full
Constant	0.172	0.260	0.280	0.248
	(0.018)	(0.016)	(0.015)	(0.015)
Firm Fixed Effects (Firm 1 = baseline):				
Firm 2	0.008	—	0.008	0.073
	(0.028)		(0.018)	(0.035)
Firm 3	0.032	—	0.032	0.054
	(0.024)		(0.016)	(0.029)
Firm 4	−0.119	—	−0.119	−0.077
	(0.023)		(0.014)	(0.021)
Position Fixed Effects (Position A = baseline):				
Position B	—	−0.186	−0.186	−0.205
		(0.019)	(0.015)	(0.017)
Position C	—	−0.123	−0.123	−0.071
		(0.023)	(0.015)	(0.022)
Position D	—	−0.124	−0.124	−0.026
		(0.026)	(0.018)	(0.023)
Observations	96	96	96	96
R^2	0.328	0.434	0.762	0.878
Adjusted R^2	0.306	0.415	0.746	0.855
F-statistic	18.72	30.52	49.78	93.95

Notes: Dependent variable is improvement in selection probability. Robust standard errors (HC3) in parentheses.

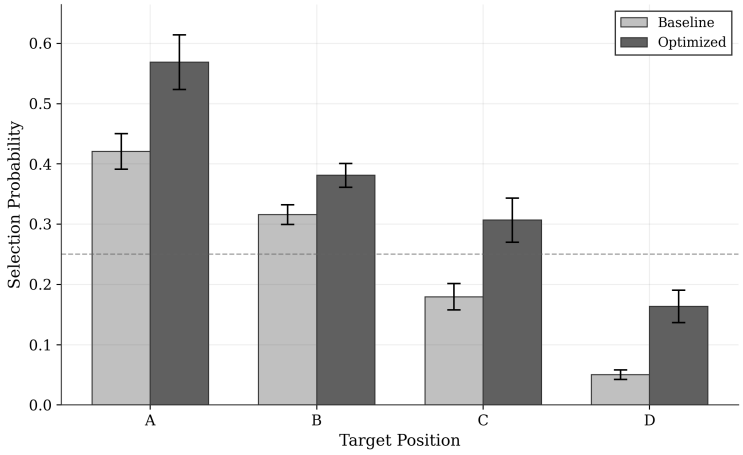


Figure 4: Treatment effect by position: Baseline versus optimized selection probabilities across target positions A, B, C, and D with standard error bars.

As shown in Figure 4, position A achieves 26.0% baseline improvement compared to only

7.4%, 13.7%, and 13.6% for positions B, C, and D respectively. This aligns with documented position bias in LLMs, where earlier options receive preferential attention.

Model (1) with firm fixed effects alone explains 32.8% of the variance, with Firm 4 showing significantly lower responsiveness (-11.9 percentage points, $p = 0.001$). Figure 5 visualizes these firm-specific differences.

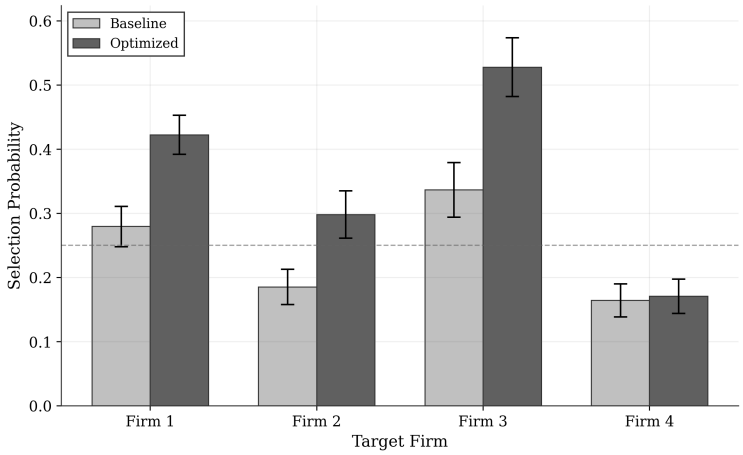


Figure 5: Treatment effect by firm: Baseline versus optimized selection probabilities across Firms 1-4 with standard error bars.

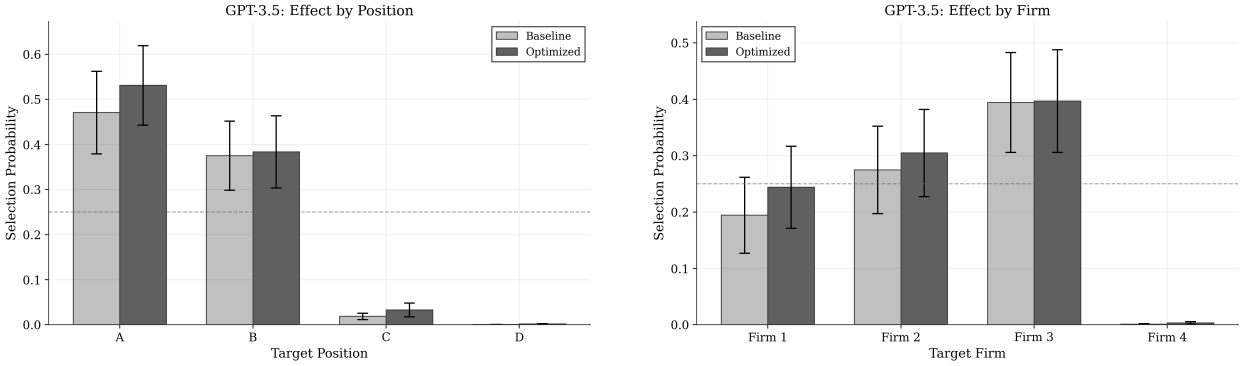
As Figure 5 demonstrates, while Firms 1, 2, and 3 achieve relatively similar optimization levels, Firm 4 shows markedly lower responsiveness to optimization. The two-way fixed effects model (3) dramatically improves explanatory power ($R^2 = 0.762$), and the full model (4) with interactions achieves $R^2 = 0.878$.

Cross-Model Generalization: GPT-3.5-turbo

To assess cross-model generalization, we start by validating all optimized sequences on OpenAI GPT-3.5-turbo, which represents one of the largest commercially available model that exposes output log probabilities through its API. With approximately 175B parameters (20× larger than LLaMA-3), GPT-3.5-turbo allows us to directly measure $P(e_{\text{target}}|\mathbf{x}^*, \mathcal{M}_{GPT})$ for our optimized sequences $\mathbf{x}^*$ without requiring gradient access.

The cross-model validation revealed limited transfer of optimization effects. While LLaMA-

3 showed robust improvements averaging 11.3 percentage points, GPT-3.5 exhibited only marginal gains of 2.1 percentage points (from 21.6% to 23.7%), which did not reach conventional statistical significance ($t = 1.84$, $p = 0.069$, Cohen's $d = 0.188$).



(a) Effect by Position **(b)** Effect by Firm

Figure 6: GPT-3.5 transfer results revealing extreme position bias. Error bars represent standard errors. The dashed line indicates random chance (25%). Note the striking baseline distribution: positions A and B dominate with 88% combined probability, while positions C and D receive near-zero attention. This extreme position bias creates a ceiling effect for positions already preferred and a floor effects for neglected positions.

Critically, Figure 6 reveals that GPT-3.5 exhibits far stronger position biases than LLaMA. The baseline probabilities show extreme concentration in positions A (47.1%) and B (37.5%), with positions C and D receiving virtually no consideration (2.4% and 0.3% respectively). This severe position bias—where 88% of baseline probability concentrates in the first two positions—fundamentally constrains the optimization's ability to transfer. Position A, already near saturation, shows modest gains (+6.0 percentage points), while positions C and D, despite substantial improvements in LLaMA-3, remain trapped near zero due to GPT-3.5's overwhelming bias against later positions.

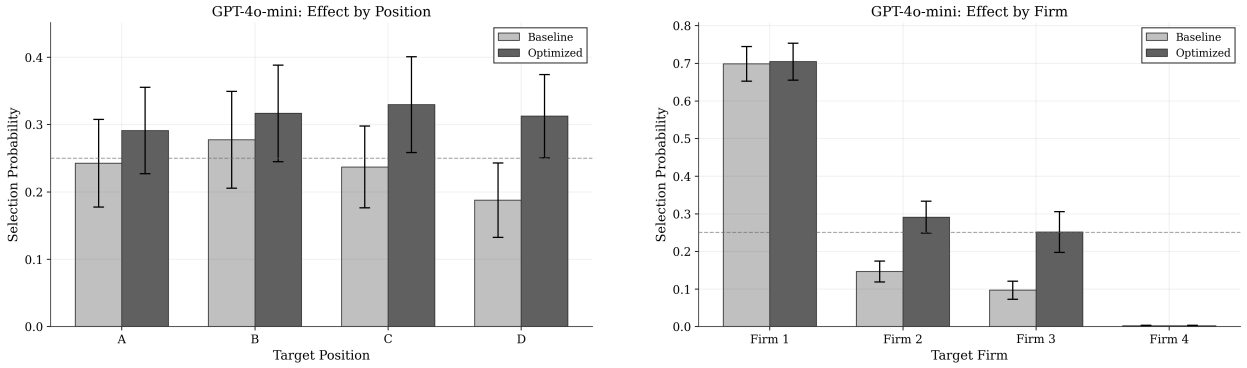
This pattern suggests that some models may develop stronger structural biases that resist manipulation through semantic optimization. The limited transfer is thus not merely a function of model scale (8B vs 175B parameters) but reflects qualitatively different attention mechanisms, where GPT-3.5's extreme position preferences create both ceiling effects for favored positions and floor effects for disfavored ones, effectively nullifying optimizations

that successfully redistribute attention in more balanced models like LLaMA-3.

Empirical Validation: GPT-4o-mini

For our final validation, we employ GPT-4o-mini, representing the frontier of commercially deployed LLMs. Both GPT-4o-mini and GPT-3.5-turbo are accessible only via proprietary APIs without model weights, offering at most limited access to internal inference signals and reflecting the trend toward increasingly closed architectures in state-of-the-art systems. This opacity necessitates an empirical sampling approach that we illustrate with this model: we query the model $N = 100$ times with temperature $T = 1.2$ to construct empirical distributions of model outputs for each of our experiments with the initial and final optimized text. While computationally expensive (we ran 9600 total inferences on the model), this Monte Carlo approach allows us to estimate $\hat{P}(e_{\text{target}}|\mathbf{x}, \mathcal{M}_{\text{GPT-4o}})$ through frequency counts, providing confidence intervals via $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$.

Remarkably, GPT-4o-mini demonstrates robust transfer of optimization effects, achieving a statistically significant mean improvement of 7.6 percentage points (from 23.6% to 31.2%, $t = 4.99$, $p = 0.001$, Cohen's $d = 0.51$). This successful transfer to a more advanced and larger model contrasts sharply with the limited transfer observed in GPT-3.5, suggesting that GPT-4o's more balanced architecture may be more receptive to semantic optimization.



(a) Effect by Position **(b)** Effect by Firm

Figure 7: GPT-4o-mini transfer results showing successful generalization. Error bars represent 95% confidence intervals from empirical sampling (N=100 each experiment, 9600 total calls). The dashed line indicates random chance (25%).

As shown in Figure 7, GPT-4o-mini exhibits markedly different transfer patterns compared to GPT-3.5. While GPT-3.5's extreme position bias (88% probability concentrated in positions A and B) created insurmountable ceiling and floor effects, GPT-4o's more balanced baseline distribution enables meaningful optimization transfer across all positions. Positions C and D, which were effectively unreachable in GPT-3.5, show improvements of 9.3 and 12.5 percentage points, respectively, in GPT-4o.

The firm-level analysis reveals strong heterogeneity in transfer effectiveness. Firms 2 and 3 show highly significant improvements (13.9% and 14.9% respectively, both $p = 0.01$), while Firm 4 remains unresponsive—consistent across all tested models. The firm effects explain 24.4% of variance (Model 1), substantially more than position effects (5.3%), suggesting that certain firm descriptions are inherently more amenable to optimization regardless of position.

This successful transfer to GPT-4o-mini, despite its opacity and larger scale, demonstrates that this optimization framework can influence even state-of-the-art models when they lack the extreme structural biases present in GPT-3.5. The empirical validation confirms that constrained semantic modifications remain effective against completely opaque models.

DISCUSSION

Our findings highlight an important asymmetry in the effectiveness of constrained content optimization. When optimization is evaluated on the same model that guides the edits (e.g., an open model such as LLaMA-3-8B), improvements are consistently large, statistically significant, and robust across firms and positions. In this within-model setting, constrained optimization systematically enhances selection probabilities, confirming that the method identifies semantically meaningful edits that increase a firm's likelihood of recommendation while maintaining content coherence.

However, cross-model evaluation reveals that these gains do not always generalize uniformly. While GPT-4o-mini exhibits substantial and statistically significant improvements,

transfer to GPT-3.5-turbo is far more limited. This divergence cannot be explained by scale alone, but rather reflects structural differences in attention allocation. Specifically, GPT-3.5 displays an extreme position bias, with nearly ninety percent of its baseline probability mass concentrated in the first two positions. In such cases, semantic improvements are attenuated by a ceiling effect for early positions and a floor effect for later ones, even when the optimized content is more compelling. These results suggest that semantic validity remains intact across models, but the expression of this meaning is mediated by model-specific structural biases.

This structural bias has important implications for marketing practice. Given that position effects in LLMs likely arise from retrieving pages from search engine results, our results show that effective SEO helps with LLM visibility optimization through this position effect (Chen et al. 2025). Our findings suggest that LLMs may sustain rather than attenuate positional inequalities that arise from search. At the same time, for firms appearing lower in search results, our content optimization approach provides an opportunity to neutralize the advantages competitors have in SEO.

While this evaluation is experimental, the observed changes in LLM recommendation probability through content optimization imply meaningful shifts in firm visibility within AI-mediated discovery, which will persist with more sophisticated techniques. In marketing terms, this effect is similar to gaining several ranks in search engine results, which can influence awareness, engagement, and conversion. Even without behavioral data, these results quantify a mechanism with clear economic relevance.

Taken together, these results extend the SEO literature into the domain of generative engines. They show that while constrained content optimization identifies semantic patterns with genuine cross-model validity, the degree to which such optimizations translate into improved visibility depends on how firms already rank in search results and the extent to which foundational models demonstrate strong position preferences. In practice, this means that optimization strategies must be evaluated not only for their semantic coherence and

authenticity, but also for their interaction with the attention biases of the specific LLMs.

Managerial Implications

The emergence of LLMs as intermediaries in consumer decision-making fundamentally alters how firms must approach market visibility. Organizations now optimize for dual audiences: human consumers seeking authentic information and AI systems that mediate discovery through opaque inference mechanisms. This work demonstrates that this challenge is both tractable and economically consequential.

The expanding ecosystem of open-source models, from Meta's LLaMA, DeepSeek, and Google's Gemma to OpenAI's recent release of 20B and 120B parameter models, creates unprecedented opportunities for optimization exploration. These models function as optimization laboratories where firms can identify semantic patterns that transfer to opaque commercial systems like GPT-4 and Claude. Our finding that optimizations can generalize across architectures suggests that insights gained from accessible open models have broader strategic value, providing a methodological foundation for influencing AI systems that increasingly shape market outcomes.

The pronounced position effects we document carry immediate strategic implications. Firms consistently relegated to later positions in AI recommendations can potentially experience improvements exceeding 30 percentage points, whereas those already favored by position bias encounter ceiling effects that limit the returns of optimization.

Perhaps most significantly, this heterogeneity necessitates position-aware strategies: organizations must first diagnose their typical AI ranking before allocating optimization resources. We analyze the interaction between generative and search enhancement for LLM recommendations. Our constrained vocabulary approach exemplifies a reconciliation of AI optimization and content authenticity. By restricting modifications to professionally appropriate terminology, firms can enhance AI visibility without compromising credibility with human audiences. As AI systems increasingly mediate commercial discovery, the ability to

influence these systems while maintaining professional integrity will likely transition from specialized technical skill to core marketing capability. The current low adoption rates suggest a closing window of opportunity for competitive advantage through early implementation.

Authenticity, Policy and Ethical Considerations

We present an instance of a commercially viable technique that maintains authenticity and factual accuracy, semantic coherence, and brand consistency while improving AI visibility. By constraining edits to professional vocabulary, the method avoids manipulative or deceptive text changes. In practice, image-to-text representations may differ from the designer's intended narrative, making careful optimization of captions and alt-text strategically important. Such practices must also respect accessibility standards such as the Web Content Accessibility Guidelines (WCAG) (World Wide Web Consortium (W3C) 2023), which require that images containing informational content include equivalent text alternatives for users who are blind or have low vision. By adhering to these principles, firms can enhance AI discoverability while ensuring transparency, inclusivity, and integrity in digital communication.

However, we point out that the same mechanism can be exploited for manipulative or misleading applications. Optimizing descriptions to maximize AI-driven visibility could incentivize firms to deviate from authentic representations of their products or services, emphasizing algorithmic relevance over factual correspondence. Such practices may distort consumer perception, undermine trust in AI-mediated recommendations, and erode the informational integrity of generative search ecosystems.

To prevent similar dynamics in generative environments, firms and regulators must establish clear standards that distinguish permissible optimization from manipulative practices. Transparency in AI-directed content adjustments, adherence to accessibility and factuality norms, and third-party auditing mechanisms can help preserve the credibility of AI-mediated

markets. Ethical generative engine optimization requires aligning algorithmic visibility with truthful representation, ensuring that efforts to improve AI discoverability do not compromise consumer welfare or professional integrity.

CONCLUSIONS

This study demonstrates that constrained vocabulary optimization can meaningfully influence LLM recommendation behavior while maintaining professional authenticity and semantic coherence. Our framework achieved statistically significant improvements on Meta LLaMA-3-8B, increasing target firm selection probability by 15.2 percentage points (from 26.8% to 42.0%), representing a 56.7% relative improvement. This within-model optimization proved robust across all 96 experimental configurations spanning every possible firm-position combination.

We found cross-model generalization revealed the role of positional biases in determining optimization effectiveness. While optimization successfully transferred to GPT-4o-mini (7.6 percentage point improvement, $p = 0.001$), transfer to GPT-3.5-turbo was severely attenuated (2.1 percentage points, $p = 0.069$). This divergence cannot be explained by model scale alone, particularly GPT-3.5-turbo's position bias, which concentrates 88% of baseline probability in positions A and B, creating insurmountable ceiling and floor effects that resist semantic optimization. In contrast, GPT-4o-mini's more balanced attention distribution enables meaningful improvements across all positions.

Our findings extend the search engine optimization (SEO) literature into the era of generative AI, revealing that optimization effectiveness depends critically on the interaction between semantic modifications and model-specific architectural biases. Position effects dominated variance in our experiments, with position A achieving 26.0 percentage point improvements compared to 7.4%, 13.7%, and 13.6% for positions B, C, and D, respectively. This suggests that LLMs may amplify rather than mitigate the positional inequalities documented in traditional search contexts.

Unlike adversarial attacks that generate detectable anomalies, our constrained approach maintains professional authenticity through domain-appropriate vocabulary restrictions and coherence filtering. The successful transfer to state-of-the-art models like GPT-4o-mini, despite complete opacity to gradient information, demonstrates that our framework identifies semantic patterns with genuine cross-architectural validity. As LLMs increasingly mediate commercial information flow, this methodology offers organizations an ethical pathway to influence AI recommendations without compromising content integrity, while also providing a complementary technique to their SEO efforts (if any) at comparable cost.

LIMITATIONS AND FUTURE WORK

Several important limitations constrain the interpretation and generalizability of our findings. First, our experiments rely on controlled simulations rather than field deployment. While we demonstrate effectiveness in laboratory conditions with authentic architectural firm content, we lack evidence of real-world impact on actual consumer behavior, web traffic, or business outcomes. Future research should conduct randomized field experiments measuring click-through rates, engagement metrics, and conversion when optimized descriptions are deployed in production LLM systems.

Second, our optimization horizon was limited to 30 iterations with 8-token descriptions. Extended optimization runs might reveal different convergence patterns, diminishing returns, or potential overfitting to specific model architectures. Similarly, longer descriptions common in commercial contexts may exhibit different optimization dynamics than our constrained experimental setup.

Third, ethical considerations merit deeper examination. While our constrained approach maintains semantic coherence, widespread adoption could create an "optimization arms race" that degrades information quality. Future work should investigate detection methods, establish best practices for responsible optimization, and examine the societal implications of AI-mediated information filtering.

Finally, our focus on architectural firms represents a specific use case where visual assets must be conveyed through textual metadata. Different domains with varying content characteristics, such as e-commerce products, academic papers, or news articles, may exhibit distinct optimization patterns. Domain-specific studies would reveal whether our findings generalize beyond visually-oriented professional services.

Despite these limitations, our framework's demonstrated effectiveness across architecturally distinct models, combined with its maintenance of professional authenticity, suggests practical applicability for organizations navigating the emerging landscape of AI-mediated markets. As LLMs increasingly shape information discovery and commercial recommendations, understanding how to ethically influence these systems while preserving content integrity becomes critical for both researchers and practitioners.

Although demonstrated with architectural firms, the mechanism operates on text-embedding geometry rather than domain-specific content. Therefore, similar constrained optimizations should apply to other descriptive domains, such as e-commerce catalogs, hospitality listings, and professional services, where AI systems index textual metadata. Future work should test these cross-domain extensions empirically.

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APPENDIX

FIRM SELECTION PROMPT

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>
You are evaluating firms for a Public Library. You must select the BEST firm based on
    the provided information.<|eot_id|><|start_header_id|>user<|end_header_id|>
Search results:
A) [REDACTED] Metropolitan Library: [REDACTED] Branch by [REDACTED]
   Image: A cantilevered glass box floating above the street
   Description: The 24,500-square-foot [REDACTED] Library reimagines libraries as
    community catalysts, balancing quiet reading spaces with dynamic social areas. The
    design features a dramatic cantilevered reading loft that engages the street,
    providing elevated city views while sheltering a public plaza below. The facility
    serves as a civic anchor for the surrounding urban neighborhood.
B) City of [REDACTED] Public Library by [REDACTED]
   Image: A curved glass pavilion with accessible parking plaza
   Description: The [REDACTED]-designed renovation and expansion of [REDACTED] Public
    Library transforms the existing building into a light-filled community hub featuring
    a curved-glass entry pavilion, geothermal heating and cooling, and flexible
    multipurpose spaces. The renovation rather than rebuild approach reduced
    environmental impact while preserving the library's role as a civic cornerstone.
C) [REDACTED] Public Library by [REDACTED]
   Image: Professional workspace modern design collaborative
   Description: The [REDACTED] Public Library transforms a former rocket factory into a
    vibrant community hub celebrating the Blackland Prairie region. As Texas's sixth
    largest library, it prioritizes flexibility with all bookshelves on wheels for
    adaptable configurations. The space promotes collaboration, creativity, and
    innovation, serving as a modern community center beyond traditional book lending.
D) [REDACTED] Library by [REDACTED]
   Image: A colorful striped facade with terracotta vertical screens
   Description: The [REDACTED] Library serves [REDACTED]'s diverse Neighbourhood
    Improvement Area within a walled garden defined by a polychrome terracotta screen.
    Three glazed courtyards organize distinct zones for children, teens, and adults,
    bringing light and nature throughout. Built on an adjacent parking lot, the existing
    library remained open during construction, ensuring continuous community service.
Which firm would be the BEST choice? Respond ONLY with a single character of your
    choice, no explanation or any other text, just one letter.
The best choice is:<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

VOCABULARY BAG PROMPT:

```
Analyze this architectural project image and description.
Extract ALL possible terms that could be used to describe
this specific project.
Include:

1. Visible architectural elements (materials, forms, spaces)
2. Specific features mentioned in the description
3. Technical specifications (square footage, numbers, dimensions)
4. Location/context references
```

```
5. Program elements (caf, meeting rooms, etc.)
6. Design concepts stated or visible
7. Any unique or distinguishing features

Output a comprehensive list of individual TOKENS (single words only)
that are factually accurate for THIS project only.
Include both generic architectural terms that apply AND
specific elements unique to this project.

Do not include:
- Terms that COULD apply but aren't visible or mentioned
- Generic praise words (innovative, beautiful, etc.)
- Multi-word combinations

Format: One token per line
```